Cloud Computing 101

Introduction to Cloud Computing, Part 2

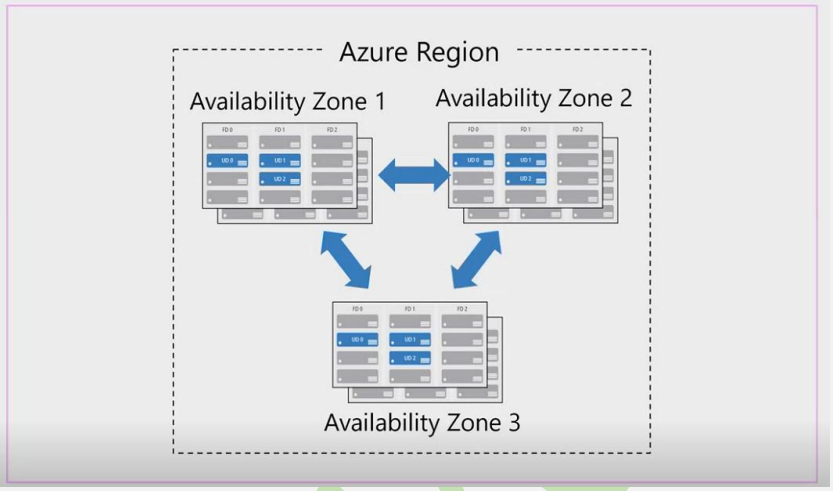
Azure Regions

Cloud regions are groups of nearby data centers designed to deliver low latency, scalability, and compliance, which is a key foundation of cloud computing. This documentation focuses on **Azure regions** and their role in cloud infrastructure.

Availability zones are physically separate datacenters within the same region, designed to improve reliability and protect workloads from datacenter-level failures.

Availability zones

- Provide protection against downtime due to datacenter failure.
- Physically separate datacenters within the same region.
- Each datacenter is equipped with independent power, cooling, and networking.
- Connected through private fiber-optic networks.



Azure region pairs for disaster recovery and high availability

Region pairs

- At least 300 miles of separation between region pairs.
- Automatic replication for some services.
- Prioritized region recovery in the event of outage.
- Updates are rolled out sequentially to minimize downtime.
- Web link: https://aka.ms/PairedRegions

Region	Region
North Central US	South Central US
East US	West US
West US 2	West Central US
US East 2	Central US
Canada Central	Canada East
North Europe	West Europe
UK West	UK South
Germany Central	Germany Northeast
South East Asia	East Asia
East China	North China
Japan East	Japan West
Australia Southeast	Australia East
India South	India Central
Brazil South (Primary)	South Central US

Azure sovereign regions are isolated cloud environments designed to meet strict government security and compliance requirements

Azure sovereign regions (US government services)

Meets the security and compliance needs of US federal agencies, state and local governments, and their solution providers.

Azure government:

- Separate instance of Azure.
- Physically isolated from non-US government deployments.
- Accessible only to screened, authorized personnel.

Azure China is a sovereign cloud environment designed to meet China's regulatory and data residency requirements

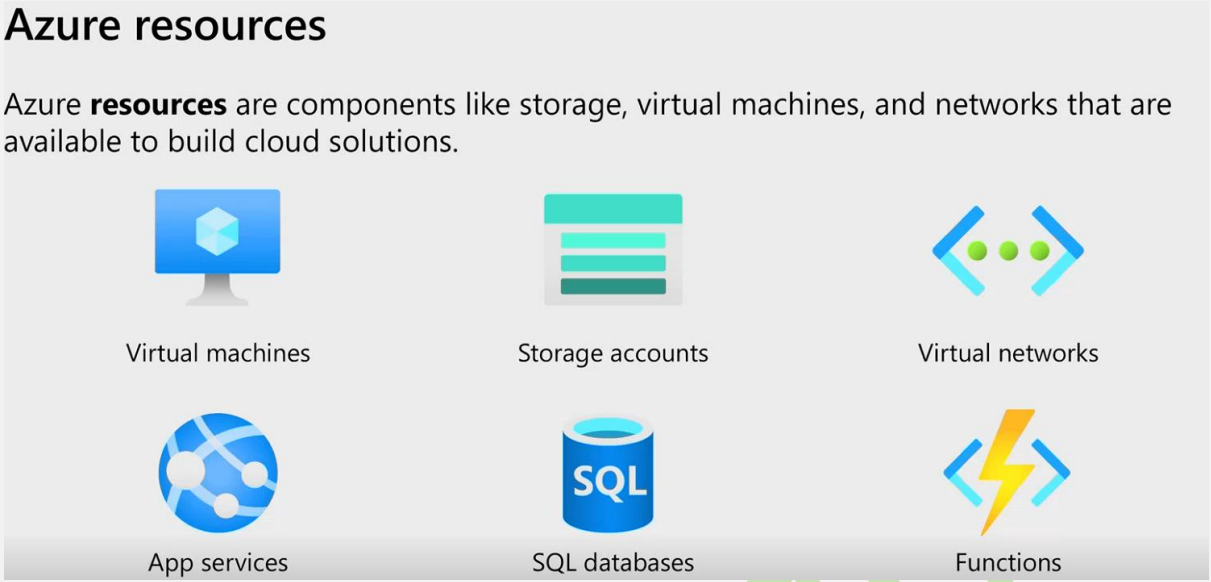
Azure sovereign regions (Azure China)

Microsoft is China's first foreign public cloud service provider, in compliance with government regulations.

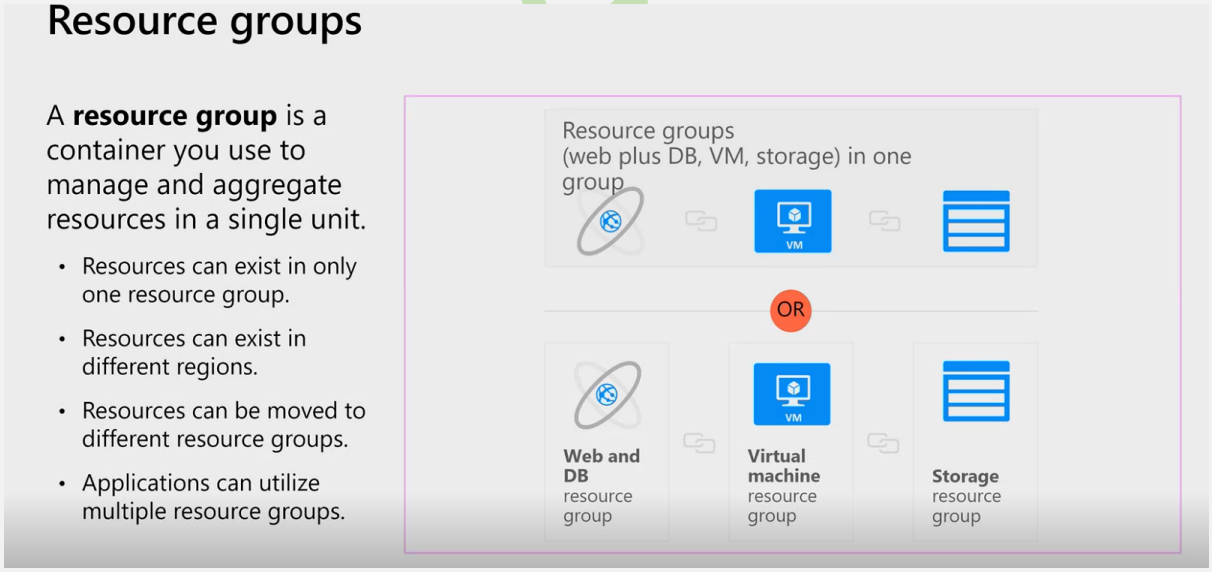
Azure China features:

- Physically separated instance of Azure cloud services operated by 21Vianet.
- All data stays within China to ensure compliance.

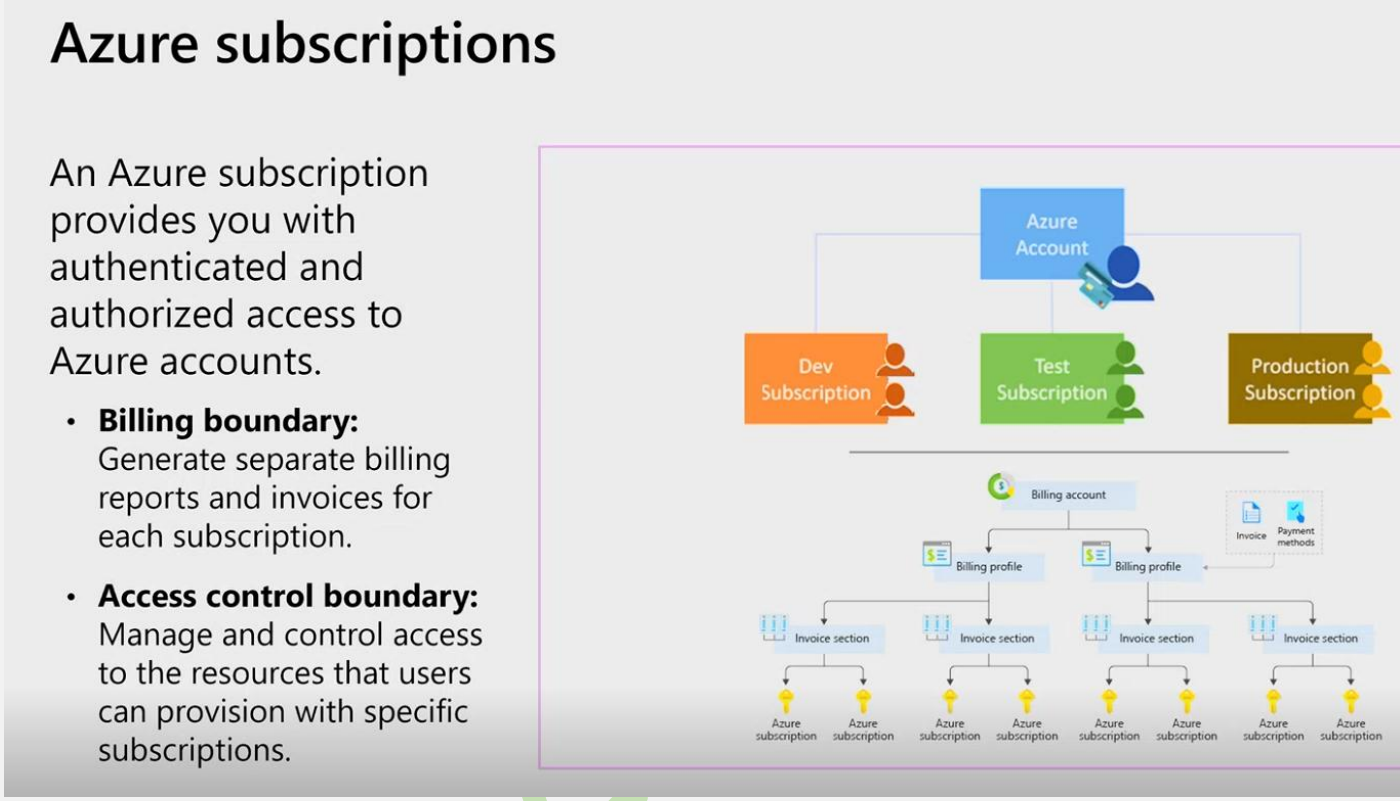
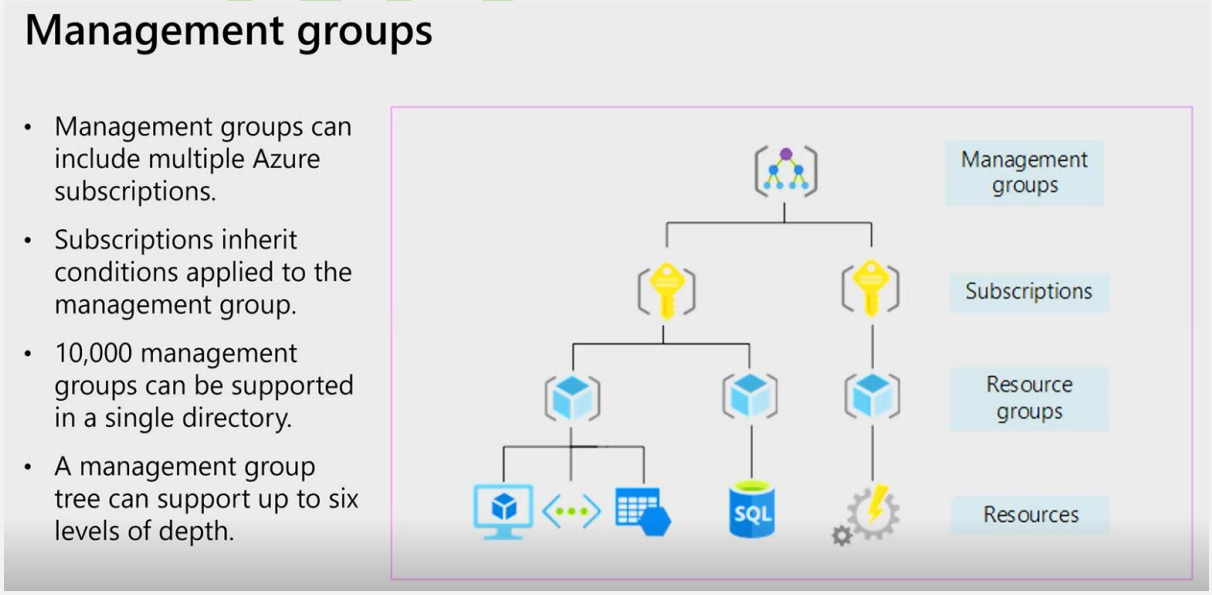
Essential building blocks for constructing and deploying cloud-based solutions on Azure.



Azure Resource Groups: Efficiently manage and organize cloud resources for seamless application deployment



Understanding Azure subscriptions: Key boundaries for billing and access control in a multi-environment setup



Organizing Azure resources through management groups, subscriptions, and resource groups for efficient cloud governance.